Building the DE-9IM Matrix

GIScience and Geodatabases — Vector Data

2026-10-30

Name: _____ **Date:** _____

Reference Card

Every geometry has three disjoint parts:

Part	Polygon	Line	Point
Interior (I)	The area inside (not the edge)	The line, excluding endpoints	The point itself
Boundary (B)	The exterior ring(s)	The two endpoints	*empty*
Exterior (E)	Everything outside	Everything outside	Everything except the point

For each pair of geometries, fill in the **DE-9IM matrix** — the dimension of each part-pair intersection:

F = empty **0** = point **1** = line **2** = area

	I(B)	B(B)	E(B)
I(A)			
B(A)			
E(A)			

Read the 9 values left-to-right across rows → the **DE-9IM string** (e.g. `"212101212"`).

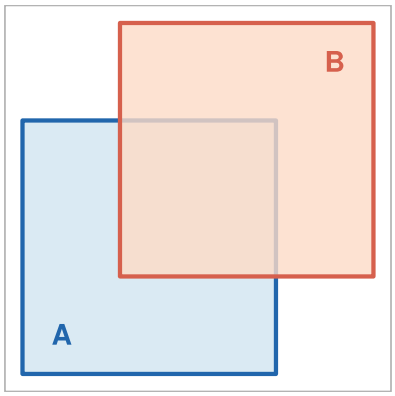
Worked example — two completely separate polygons:

I(A)∩I(B) = **F**, I(A)∩B(B) = **F**, I(A)∩E(B) = **2** (A's interior lies entirely in B's exterior) B(A)∩I(B) = **F**, B(A)∩B(B) = **F**, B(A)∩E(B) = **1** (A's boundary is a line in B's exterior) E(A)∩I(B) = **2**, E(A)∩B(B) = **1**, E(A)∩E(B) = **2**

→ String: **`FF2FF1212`** → predicate: `disjoint`

Instructions: For each scenario, fill in the 3×3 matrix, write the DE-9IM string, and name the predicate.

Scenario 1



A and B overlap partially; neither contains the other.

	I(B)	B(B)	E(B)
I(A)			
B(A)			
E(A)			

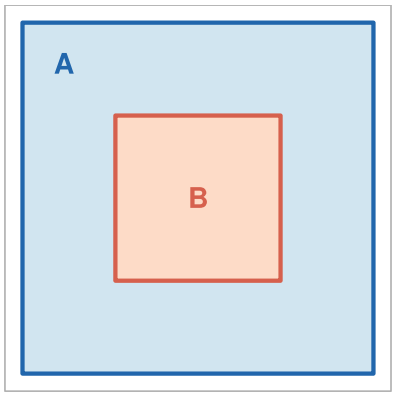
String: _ _ _ _ _ _ _ _ _

Predicate: _____

Which forests extend into the canton of Luzern?

_____

Scenario 2



B lies completely inside A; their boundaries do not touch.

	I(B)	B(B)	E(B)
I(A)			
B(A)			

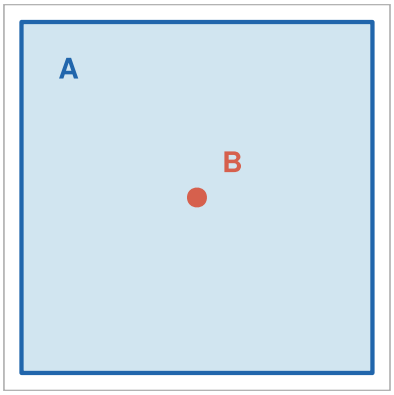
	I(B)	B(B)	E(B)
E(A)			

String: _ _ _ _ _ _ _ _ _

Predicate: _____

Is the building footprint entirely within the flood zone?

Scenario 3



B is a point strictly inside polygon A (not on the edge).

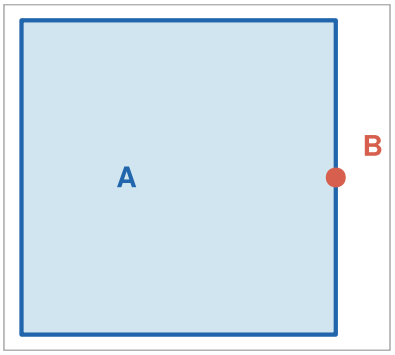
	I(B)	B(B)	E(B)
I(A)			
B(A)			
E(A)			

String: _ _ _ _ _ _ _ _ _

Predicate: _____

Note: what is B(B) for a point? How does this affect the second column?

Scenario 4



B is a point exactly on the boundary of polygon A.

	I(B)	B(B)	E(B)
I(A)			
B(A)			
E(A)			

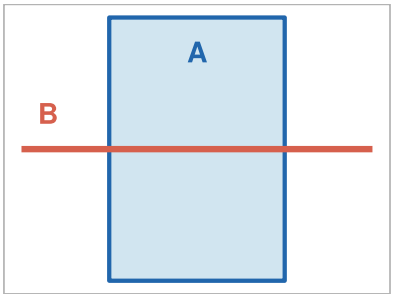
String: _ _ _ _ _ _ _ _ _

Predicate: _____

Compare with Scenario 3: which cells change, and why?

_____

Scenario 5



Line B enters and exits polygon A; both endpoints of B lie outside A.

	I(B)	B(B)	E(B)
I(A)			
B(A)			

	I(B)	B(B)	E(B)
E(A)			

String: _ _ _ _ _ _ _ _ _

Predicate: _____

Note: B(B) are the line's endpoints. Where do they fall relative to A?

Bibliography